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United States Patent	12396908
Kind Code	B2
Date of Patent	August 26, 2025
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Step stool and method of use

Abstract

A bed step stool for assisting a user into and out of bed comprising a first support; a second support; a wheel assembly coupled to a side of the bed step stool; a step assembly supported between the first support and the second support and enabling the user to step thereon for assisting the user into and out of the bed; wherein the bed step stool includes a first orientation where the bed step stool is stabilized on the surface and the wheel assembly coupled to the side of the bed step stool is out of the way, and a second orientation where the bed step stool is tilted towards the side and is supported by the wheel assembly coupled to the side of the bed step stool, allowing rolling movement of the bed step stool on the surface via the one or more wheels of the wheel assembly.

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Family ID:	1000008782207
Appl. No.:	18/525000
Filed:	November 30, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240091086 A1	Mar. 21, 2024

Related U.S. Application Data

continuation parent-doc US 15175872 20160607 US 10034807 20180731 child-doc US 16042930
continuation-in-part parent-doc US 17723282 20220418 US 11850197 child-doc US 18525000
continuation-in-part parent-doc US 17087448 20201102 US 11311437 child-doc US 17723282
continuation-in-part parent-doc US 16782410 20200205 US 11083655 20210810 child-doc US 17087448

continuation-in-part parent-doc US 16372181 20190401 US 10835432 20201117 child-doc US 16782410
continuation-in-part parent-doc US 16262668 20190130 US 10456310 20191029 child-doc US 16372181
continuation-in-part parent-doc US 16042930 20180723 US 10213353 20190226 child-doc US 16262668
us-provisional-application US 62184102 20150624

Publication Classification

Int. Cl.: **A61G7/053** (20060101); **A47C12/00** (20060101)

U.S. Cl.:

CPC **A61G7/053** (20130101); **A47C12/00** (20130101);

Field of Classification Search

CPC: A61G (7/053); A61G (2203/36); A61G (7/1003); A61G (7/1006); A47C (12/00); A61H (2003/005); A61H (2003/004)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/723,282, filed Apr. 18, 2021, which is a continuation-in-part of U.S. patent application Ser. No. 17/087,448, filed Nov. 2, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/782,410, filed Feb. 5, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/372,181, filed Apr. 1, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 16/262,668, filed Jan. 30, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 16/042,930, filed Jul. 23, 2018, which is a continuation of U.S. patent application Ser. No. 15/175,872 filed Jun. 7, 2016, which claims the benefit of provisional patent application No. 62/184,102 filed on Jun. 24, 2015. All of these prior applications are incorporated by reference herein.

FIELD OF THE INVENTION

(1) The present subject matter relates generally to a step stool for entry to a bath tub. More specifically, the present invention relates to a step stool and guardrail that allows users to get into a bathtub from the side of the bathtub by way rails set at a perpendicular angle to the bathtub.

BACKGROUND OF THE INVENTION

(2) Seniors and individuals with limited mobility must employ a variety of protective measures when it comes to simple, everyday tasks such as getting in and out of bathtub, bed, and even sleeping. They may use bed rails running parallel to the bed in order to prevent falls, and step stools to help them get in to bed.

(3) A difficulty with bed rails is that they are usually attached to the bed, typically by being placed between the mattress and box spring of the bed. They are not mobile or portable. This makes the user's sleeping situation less flexible.

(4) Additionally, even using a combination of bed rail and step stool, seniors and persons with limited mobility are still required to enter their bed from the top or the bottom (where there are no guardrails). This may inconvenience individuals who must climb up their beds or over their pillows in order to arrive at their desired resting position.

(5) Another challenge for seniors and individuals with limited mobility is getting in and out of a

bathub. The high walls of the bathtub, along with a possibly wet/slippery surface of a bathtub and/or bathroom floor, can make getting into and out of a bathtub treacherous for seniors and individuals with limited mobility

SUMMARY OF THE INVENTION

(6) An aspect of the disclosure involves a bed step stool for assisting a user into and out of bed including opposite sides, a head, and a foot comprising a first support to be disposed closer to the head of the bed; a second support to be disposed closer to the foot of the bed, the bed step stool including a side and at least one of the first support and the second support including an upper portion configured to be gripped by a user; a wheel assembly coupled to the side of the bed step stool and including one or more wheels configured to roll on a surface; a step assembly supported between the first support and the second support, the step assembly to enable the user to step thereon for assisting the user into and out of the bed; wherein the first support and the second support define an unobstructed pathway for entry and exit with respect to the bed and the bed step stool that is perpendicular with respect to the sides of the bed whereby the user goes between the first support and the second support, onto the step assembly, and off the step assembly without any obstruction, and the bed step stool including a first orientation where the bed step stool is stabilized on the surface and the wheel assembly coupled to the side of the bed step stool is out of the way, and a second orientation where the bed step stool is tilted towards the side and is supported by the wheel assembly coupled to the side of the bed step stool, allowing rolling movement of the bed step stool on the surface via the one or more wheels of the wheel assembly.

(7) One or more implementations of the aspect of invention described above includes one or more of the following: the wheel assembly is detachably coupled to the side of the bed step stool.

(8) Another aspect of the disclosure involves a bed step stool for assisting a user into and out of bed including opposite sides, a head, and a foot comprising: a first support to be disposed closer to the head of the bed; a second support to be disposed closer to the foot of the bed; a wheel assembly coupled to the side of the bed step stool and including one or more wheels configured to roll on a surface; a step assembly supported between the first support and the second support, the step assembly to enable the user to step thereon for assisting the user into and out of the bed; a raising and lowering mechanism disposed under the step assembly configured to raise and lower the step assembly; wherein the first support and the second support define an unobstructed pathway for entry and exit with respect to the bed and the bed step stool that is perpendicular with respect to the sides of the bed whereby the user goes between the first support and the second support, onto the step assembly, and off the step assembly without any obstruction, and the bed step stool including a first orientation where the raising and lowering mechanism orients the step assembly in a raised position to facilitate getting the user into and out of bed and a second orientation where the raising and lowering mechanism orients the step assembly in a lowered position to facilitate getting the user onto and off of the step relative to a support surface.

(9) One or more implementations of the aspect of invention described above includes one or more of the following: the raising and lowering mechanism is a stair stepper mechanism that lowers when the user steps on the step assembly and then recoils, lifting the user up after both feet are on the step assembly; the raising and lowering mechanism is a manual control and a hydraulic or pneumatic vertical lift mechanism that a user controls via the manual control to raise and lower the step assembly to a desired height; and/or the raising and lowering mechanism is a manual control and a counterbalance vertical lift mechanism that a user controls via the manual control to raise and lower the step assembly to a desired height.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an embodiment of the bed step stool shown in position/use adjacent to a side of a user's bed;
- (2) FIG. 2 is a perspective view of the bed step stool of FIG. 1;
- (3) FIGS. 3A, 3B, 3C, 3D, 3E, and 3F are front, rear, right, left, top, and bottom views of the bed step stool of FIG. 1;
- (4) FIG. 4A is a flow chart illustrating an exemplary method of using the bed step stool to get into the bed shown in FIG. 1;
- (5) FIG. 4B is a flow chart illustrating an exemplary method of using the bed step stool to get out of the bed shown in FIG. 1;
- (6) FIG. 5 is a perspective view of another embodiment of a bed step stool;
- (7) FIG. 6A is a perspective view of a further embodiment of a bed step stool;
- (8) FIG. 6B is a perspective view of a still further embodiment of a bed step stool;
- (9) FIG. 7 is a perspective view of a step stool that may be used with a bathtub;
- (10) FIGS. 8A, 8B, 8C, 8D are perspective views of a further embodiment of a bed step stool shown attached to a bed, but with different embodiments of a bed attachment shown;
- (11) FIG. 9 is a perspective view of a bathtub step stool shown attached to a bathtub;
- (12) FIG. 10 is a rear perspective view of the bathtub step stool of FIG. 9;
- (13) FIG. 11 is another rear perspective view of the bathtub step stool of FIG. 9;
- (14) FIG. 12 is a front elevational view of the bathtub step stool of FIG. 9;
- (15) FIG. 13 is a left side elevational view of the bathtub step stool of FIG. 9;
- (16) FIG. 14 is a rear elevational view of the bathtub step stool of FIG. 9;
- (17) FIG. 15 is a top plan view of the bathtub step stool of FIG. 9;
- (18) FIG. 16A is a perspective view of another embodiment of a step stool shown attached to a bathtub;
- (19) FIG. 16B is a side-elevational view of area 16B of a glow-in-dark strip on one the legs of the step stool of FIG. 16A;
- (20) FIG. 17 is a side-elevational view of a mount assembly of the step stool of FIG. 16A;
- (21) FIG. 18 is a perspective view of the step stool of FIG. 16A shown in a collapsed configuration;
- (22) FIG. 19 is a side-elevational view of the step stool of FIG. 16A shown in a collapsed configuration;
- (23) FIG. 20 is a perspective view of another embodiment of a bed step stool;
- (24) FIG. 21 is a perspective view of an additional embodiment of a bed step stool;
- (25) FIGS. 22A and 22B are perspective views of further embodiments of a bed step stool;
- (26) FIG. 23 is a perspective view of an additional embodiment of a bed step stool, which may also function as a walker, shown adjacent to a bed and with a seat shown in a stored position;
- (27) FIG. 24 is a perspective view of the bed step stool/walker of FIG. 23, and shows the seat in a position between the stored position and a deployed seat position;
- (28) FIG. 25 is another perspective view of the bed step stool/walker of FIG. 23, and shows the seat in a stored position;
- (29) FIG. 26 is a further perspective view of the bed step stool/walker of FIG. 23, and shows the seat in a deployed seat position;
- (30) FIG. 27 is a left side elevational view of the bed step stool/walker of FIG. 23, and shows the seat in the deployed seat position;
- (31) FIG. 28 is a rear elevational view of the bed step stool/walker of FIG. 23, and shows the seat in the deployed seat position;
- (32) FIG. 29 is a top plan view of the bed step stool/walker of FIG. 23, and shows the seat in the deployed seat position;
- (33) FIG. 30 is a perspective view of another embodiment of a bed step stool, which may also

- function as a walker, shown adjacent to a bed and with a seat shown in a stored position;
- (34) FIG. **31** is a perspective view of the bed step stool/walker of FIG. **30**, and shows the seat in the deployed seat position;
- (35) FIG. **32** is another perspective view of the bed step stool/walker of FIG. **30**, and shows the seat in the stored seat position;
- (36) FIG. **33** is a left side elevational view of the bed step stool/walker of FIG. **30**, and shows the seat in the deployed seat position;
- (37) FIG. **34** is a rear elevational view of the bed step stool/walker of FIG. **30**, and shows the seat in the deployed seat position;
- (38) FIG. **35** is a top plan view of the bed step stool/walker of FIG. **30**, and shows the seat in the deployed seat position;
- (39) FIG. **36** is a perspective view of a further embodiment of a bed step stool, which may also function as a walker, and shows a seat in a stored position;
- (40) FIG. **37** is another perspective view of the bed step stool/walker of FIG. **36**, and shows the seat in a deployed seat position;
- (41) FIG. **38** is a perspective view of a still further embodiment of a bed step stool, which may also function as a walker, and shows a securement device in a deployed securement position;
- (42) FIG. **39** is another perspective view of the bed step stool/walker of FIG. **38**, and shows the securement device in a retracted position;
- (43) FIG. **40** is a perspective view of an additional embodiment of a bed step stool, which may wheeled from location to location;
- (44) FIG. **41** is a front elevational view of another embodiment of a bed step stool, which includes a step assembly that may be raised and lowered.

DETAILED DESCRIPTION OF EMBODIMENTS OF INVENTION

- (45) Before describing a bathtub step stool, with reference to FIGS. **1** to **3F**, an embodiment of a bed step stool **100** for use by a user to assist the user to get into and out of a bed **110** will first be described. Although the bed step stool **100** is shown and described as an aid for assisting one to get into and out of a bed, the step stool **100** may be used to assist one onto and off of various types of elevated furniture. For example, but not by way of limitation, the step stool **100** may be used to assist one to get in and out of a vehicle (e.g., car, van) that may be too high for them (e.g., step stool **100** may be a different model/version having a lower cross bar in the back, but still maintain similar functionality to that shown and described herein). The step stool **100** may be used in the home to assist one onto and off of an elevated chair/stool (e.g., bar stool) taller than a kitchen chair.
- (46) The bed step stool **100** includes a long(er)/tall(er) guardrail/first support **120** parallel with and spaced with respect to a short(er) guardrail/second support **130**. The tall guardrail **120** is closer to a head **132** of the bed and the short guardrail **130** closer to a foot **134** of the bed **110**. The tall guardrail **120** is taller than a height of the bed **110** at which the user is supported when resting and the short guardrail **130** is closer in height to the height of the bed **110**. The tall guardrail **120** and the short guardrail **130** define respective parallel planes that are perpendicular with a plane define by a side **136** of the bed **110**.
- (47) A lateral crossbar assembly **140** connects (via fasteners **150**) a back leg **160** of the tall guardrail **120** to a back leg **170** of the short guardrail **130**.
- (48) Step crossbar assemblies **190**, **200** connect (via fasteners **210**) the back legs **160**, **170** to front legs **250**, **260** of the tall guardrail **120** and short guardrail **130**, respectively.
- (49) A step assembly **270** has a generous width and depth to enable a user (e.g., senior) to securely step onto the step assembly **270**. The step assembly **270** may include a slip protection tape or another slip prevention material on an upper surface. The step assembly **270** is supported between the tall guardrail **120** and the short guardrail **130** by the step crossbar assemblies **190**, **200**.
- (50) The longitudinal direction of the lateral crossbar assembly **140** and the step assembly **270** are parallel with the plane define by a side **136** of the bed **110** and perpendicular with respect to planes

defined by the tall guardrail **120** and the short guardrail **130**. When the bed step stool **100** is in position, with the crossbar assembly **140** adjacent to or abutting the side **136** of the bed **110**, the bed step stool **100** and the bed **110** form a T-shaped configuration.

(51) The tall guardrail **120** and short guardrail **130** include elevation tubes/legs **280** telescopingly received in upper tubes **320, 33, 340, 350** for adjusting the height of the entire bed step stool **100** and the height of the step assembly **270** via rotatable adjustment collars **360** and bolt holes/bolts.

(52) The step height is what goes up or down based off the bolt hole you choose (the height) that works for you. The tall and short rail's height relationship doesn't change after it is put together, only the step height and the unit as a whole then keeps the same relationship of distance.

(53) Lower ends of the elevation tubes **280** include rubber tips **400**.

(54) An upper portion **440** of the tall guardrail **120** includes an upside-down U-shaped tall grab bar **450**.

(55) An upper portion **460** of the short guardrail **130** includes an angled/inclined bar **470**.

(56) The lateral crossbar assembly **140** carries a motion-detecting lighting assembly **480** for illuminating via one or more lights (e.g., LEDs) the step assembly **270** upon detected motion (e.g., movement of the user's legs) of the user by a motion sensor.

(57) In an alternative embodiment, the bed step stool **100** includes a configuration for use with the opposite side of the bed **110**. In this alternative embodiment, instead of the tall guardrail **120** on the left side and the short guardrail **130** on the right side of the step assembly **270**, the tall guardrail **120** is on the right side and the short guardrail **130** on the left side of the step assembly **270**. The upper portions **440, 460** of the tall guardrail **120** and the short guardrail **130** (or, alternatively, the tall guardrail **120** and the short guardrail **130**) are reversible to enable the bed step stool **100** to accommodate both sides of the bed **110**.

(58) Although the bed step stool **100** shown may accommodate a wide variety of different-sized users, in an alternative embodiment, the bed step stool **100** includes one or more configurations to accommodate different types/sized users.

(59) With reference additionally to FIG. 4A, an exemplary method **500** of using the bed step stool **100** to get into the bed **110** will be described. At step **510**, the user approaches the bed step stool **100** and turns sideways towards the bed **110** and the tall grab bar **450** (i.e., user facing towards tall grab bar **450** and the head **132** of the bed **110**). As the user approaches the bed step stool **100**, the motion-detecting lighting assembly **460** illuminates the step assembly **270** upon detected movement of the user's legs. At step **520**, with hands on the tall grab bar **450**, the user steps onto the step assembly **270** and turns backside to the bed **110**. At step **530**, with hands on the tall grab bar **450**, the user sits down on the edge of the bed **110**. At step **540**, the user lays down to a resting position and lifts one's legs over the angled/inclined bar **470** of the short guardrail **130**.

(60) With reference additionally to FIG. 4B, an exemplary method **600** of using the bed step stool **100** to get out of the bed **110** will be described. At step **610**, a user reaches one's closest arm over to the tall grab bar **450**. At step **620**, the user uses the tall grab bar **450** to lift one's shoulders off of the bed **110** while swinging legs over the angled/inclined bar **470** of the short guardrail **130**, and places one's feet on the step assembly **270**. As the user swings one's legs over the angled/inclined bar **470** and one's feet approach the step assembly **270**, the motion-detecting lighting assembly **460** illuminates the step assembly **270** upon detected movement of the user's legs. At step **630**, the user takes 15-30 seconds while sitting up to gain composure. At step **640**, when the user is ready, the user uses one hand on the tall grab bar **450** (or two hands, one on tall grab bar **450**, one on angled/inclined bar **470** of the short guardrail **130**) and stands up. At step **650**, while keeping one hand on the tall grab bar **450**, the user slowly steps off the step assembly **270**, onto the floor.

(61) Advantages of the bed step stool **100** and above described methods include one or more of the following: the perpendicular orientation of the tall guardrail **120** and the short guardrail **130** with respect to of the bed step stool **100** provides a pathway to entry/exit with respect to the bed **110** whereby the user goes between the guardrails **120, 130** to step on the step assembly **270**, the bed

step stool **100** allows individuals to safely enter/exit their bed **110** from the side **136** of the bed **110**, the bed step stool **100** provides guardrails **120**, **130** for support and protection at a perpendicular angle to the longitudinal direction of the bed **110** (or plane define by the side **136** of the bed **110**), allowing the user to safely remain secure in the bed **110**, the elevation tubes/legs **280** telescopingly received in upper tubes **320**, **33**, **340**, **350** allow the user to adjust the height of the bed step stool **100** so that the bed step stool **100** can be used with a variety of bed heights and/or user heights, the bed step stool **100** allows users to enter the bed **110** at the final point of resting and eliminates the need to move the bed **110** up or down into position, the step assembly **270** of the bed step stool **100** includes a wide platform to improve safety and ease of use, the bed step stool **100** may be used in any setting, including a nursing home, a hospital, or a patient's home; the bed step stool **100** may be used with different types of beds/furniture; because the bed step stool **100** rests on legs rather than between the mattress and the box spring, there is no limit to the type of furniture/beds that the bed step stool **100** can be used in connection with, the bed step stool **100** is portable (the bed step stool **100** in the embodiment shown and described is designed not to attach to the bed, allowing one to easily store the bed step stool **100** by sliding it into a corner or closet when not in use); the bed step stool **100** helps reduce a senior's risk of falling because as they use the product their actual fear of falling is reduced; and/or the bed step stool **100** provides the user with a sense of safety (using the step **270** and grab bar(s) **450**, **470** make the user more confident in one's movements; when the user feels safer, the user reduces the risk of falling by reducing one's fear of falling).

(62) With FIG. 5, another embodiment of a bed step stool **700** that may include one or more of the below-described features will be described, with like elements to those shown and/or described with respect to FIGS. 1-4 shown and/or described with like reference numbers, but with an “a” suffix, and with the subject matter described above with respect to FIGS. 1-4 incorporated herein. Straps **710** made of hook and loop fasteners (e.g., Velcro®) or another soft material connect the lateral crossbar assembly **140a** to a bed frame to improve stability of the bed step stool **700**. Height adjustment mechanisms **720** are disposed at four corners (e.g., at bottom of guard rails **120a**, **130a** and top of legs **160a**, **170a**, **250a**, **260a**) above step assembly **730** to allow both the tall guard rail **120a** and/or the short guard rail **130a** to adjust in height in a manner different than that previously described. Step assembly **730** flips/rotates up and allows the user to have the step in a low/deployed mode/position to step on to assist in getting into and/or out of bed or retracted/high/flipped/rotated mode that allows the user to walk or step more into the middle of the bed step stool **700** without stepping up at all. Lockable wheels **740** connected to the bottom of each of the four legs **160a**, **170a**, **250a**, **260a**. The wheels **740** may be inserted in the bottom of each of the four legs **160a**, **170a**, **250a**, **260a** in lieu of height adjustment legs inserted in the bottom of each of the four legs **160a**, **170a**, **250a**, **260a**. Short guard rail **130a** raises in height to at least the same height as the tall guard rail **120a**. The tall guard rail **120a** includes a padded grip **760**.

(63) With reference to FIG. 6A, a further embodiment of a bed step stool **769** will be described. The bed step stool **769** may include one or more of the below-described features, with like elements to those shown and/or described with respect to FIGS. 1-5 are shown and/or described with like reference numbers, but with a “b” suffix, and with the subject matter described above with respect to FIGS. 1-5 incorporated herein. The tall guard rail **120b** and/or the short guard rail **130b** includes square corners **770**, **772**, **774**, **776**. Short guard rail **130b** raises in height to at least the same height as the tall guard rail **120b**. A bed rail attachment **790** rotatably attached to the tall guard rail **120b** and/or the leg **160b** serves as a bed rail to keep a user from falling/getting out of bed when the user is lying down, but swings open to allow the user to get in and out of bed easily. Although not shown, a latch or hook may be used to the bed rail attachment in the bed rail position/mode and/or the open position mode. A motor **800** operably coupled to the step assembly **270b** (e.g., via a rotatable screw mechanism) raises and lowers the step assembly **270b** vertically up and down for users (e.g., elders) who can't raise their leg to step onto the step assembly **270b** when the step assembly **270b** is raised.

(64) With reference to FIG. 6B, a still further embodiment of a bed step stool **819** will be described. Like elements to those shown and/or described with respect to FIGS. 1-6A are shown and/or described with like reference numbers, but with a “c” suffix, and with the subject matter described above with respect to FIGS. 1-6A incorporated herein. The guard rail **120c** and/or the guard rail **130c** have the same, tall height, include includes square corners **770c**, **772c**, **774c**, **776c**, and include the bed rail attachment **790c**.

(65) With reference to FIG. 7, an embodiment of a step stool **829** that may be used with a bathtub will be described. Like elements to those shown and/or described with respect to FIGS. 1-6B are shown and/or described with like reference numbers, but with a “d” suffix, and with the subject matter described above with respect to FIGS. 1-6B incorporated herein. Attachments **830** (e.g., L-shaped member that raises, lowers, rotates) that go on the back of the bath step stool **829** (e.g., attached to the tall guard rail **120d**, short guard rail **130d**, leg **160d**, leg **170d**, and/or the lateral crossbar assembly **140d**) go over the edge/side of a bathtub to sit next to the bathtub and essentially attach to the bathtub. This allows a user to set the product next to the bathtub and be stable so the individual has a step, a rail, to help people the user into and out of the bathtub. Straps **840** attach to the legs **160d**, **170d** and to bed frame attach to the bed that can help keep the step stool **829** more stable. Straps **840** made of hook and loop fasteners (e.g., Velcro®) or another soft material connect the legs **160d**, leg **170d** to a bed frame to improve stability of the step stool **700**.

(66) FIGS. 8A, 8B, 8C, 8D are perspective views of a further embodiment of bed step stools **849**, **859**, **869**, **879**, where like elements to those shown and/or described with respect to FIGS. 1-7 are shown and/or described with like reference numbers, but with an “e” suffix, and with the subject matter described above with respect to FIGS. 1-7 incorporated herein shown attached to a bed **110e**, but with different embodiments of a bed attachment **850**, **860**, **870**, **880** shown.

(67) In FIG. 8A, the bed attachment(s) **850** include strap(s) that attach to the lateral crossbar assembly **140e** (and/or the legs **160e**, leg **170e**) and run across the bed **110e** between a box spring (or other mattress support) **852** and mattress **854** and can attach to the bed frame. For example, but not by way of limitation, the bed attachment **850** may include nylon strap(s) with distal ends terminating in hook and loop fasteners (e.g., Velcro®).

(68) In FIG. 8B, the bed attachment(s) **860** are made of a solid and/or rigid material (e.g., metal, aluminum, plastics) that attach to the lateral crossbar assembly **140e** (and/or the legs **160e**, leg **170e**) and run across the bed **110e** between a box spring (or other mattress support) **852** and mattress **854** to help stabilize the bed step stool **859** when being used.

(69) In FIG. 8C, the bed attachment **870** is made of a solid and/or rigid material (e.g., metal, aluminum, plastics) that attach to the lateral crossbar assembly **140e** (and/or the legs **160e**, leg **170e**) and run across the bed **110e** between a box spring (or other mattress support) **852** and mattress **854** to help stabilize the bed step stool **869** when being used. The bed attachment **870** and/or the bed attachment **870** in connection with the lateral crossbar assembly **140e** forms a rectangular closed-loop member that runs across the bed **110e** between the box spring (or other mattress support) **852** and mattress **854**.

(70) In FIG. 8D, the bed attachment **880** is made of a solid and/or rigid material (e.g., metal, aluminum, plastics) that attach to the lateral crossbar assembly **140e** (and/or the legs **160e**, leg **170e**) and run across the bed **110e** between a box spring (or other mattress support) **852** and mattress **854** to help stabilize the bed step stool **879** when being used. The bed attachment **880** and/or the bed attachment **880** in connection with the lateral crossbar assembly **140e** forms a closed-loop member that spans most of the dimension of the mattress **854**, running across the bed **110e** between the box spring (or other mattress support) **852** and mattress **854**.

(71) In further embodiments, one or more of the step stools shown and/or described herein is a smaller version of that shown (e.g., not as wide and/or as deep as that/those shown so as to fit in smaller room).

(72) With reference to FIGS. 9-15, a further embodiment of a bathtub step stool **900** that may be

used with a bathtub **910** will be described. Like elements to those shown and/or described with respect to FIGS. **1-8D** are shown and/or described with like reference numbers, but with an “f” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-9D** incorporated herein. Attachments **920** (e.g., L-shaped members that raise, lower) that go on the back of the bath step stool **900** (e.g., attached to the leg **160f** of the first support/guard rail **120f** and is attached to the leg **170f** of the second support/guard rail **130f**) terminate at mount assembly **930**, which mounts onto/over a side/wall **940** of the bathtub **910** as shown in FIG. **9**. The mount assembly **930** includes an elongated flat bracket **950** with a pair of downwardly facing U-shaped brackets/clamps **960**. The U-shaped brackets/clamps **960** include lateral members **970** and downwardly extending members **980**. C-shaped members **990** are disposed along an underside **1000** of the lateral members **970**. The C-shaped members **990** allows the -shaped brackets/clamps **960** to be adjusted to fit the side/wall **940** of the bathtub **910** (in some parts of the bathtub **910** the side/wall **940** of the bathtub **910** is thin and other parts it's wider). A bolt may go through mount assembly **930** into C-shaped member **990** and this locks the adjustment setting in place so it's snug against side/wall **940**. The clamps **960** may be padded and may adjust to different tub wall widths. A lateral crossbar assembly **140f** is attached to the leg **160f** of the guard rail **120f** and is attached to the leg **170f** of the guard rail **130f**. The lateral crossbar assembly **140f** includes a plate **1010**. Step assembly **270f** may be vertically adjustable with respect to guard rails **120f**, **130f**. The guard rails **120f**, **130f** are of the same height and may have rubberized non slip foot/rubber tip **1012** at bottom of guard rails **120f**, **130f** and/or rubber sleeves at top of guard rails **120f**, **130f** for secure gripping of guard rails **120f**, **130f** by the user when transferring to and from the bathtub or shower. (73) In use, the bathtub step stool **900** is positioned next to the bathtub entry side/wall **940** and attached/clamped to the bathtub entry side/wall **940** via the U-shaped brackets **960**, which slide onto and over the bathtub entry side/wall **940** (and may adjust to different tub wall widths), to secure the bathtub step stool **900** in position. This allows a user to set the bathtub step stool **900** next to the bathtub **910** and be stable so the individual has a step and rails to help the user into and out of the bathtub **910**.

(74) To use the bathtub step stool **900** to enter the bathtub **910**, a user grips at least one of the first support **120f** and the second support **130f**; steps onto the step assembly **270f**, towards the entry side **940** of the bathtub **910** (passing between the first support **120f** and the second support **130f**); turns towards one of the first support **120f** and the second support **130f** while standing on the step assembly **270f**; while gripping one of the first support **120f** and the second support **130f**, steps over the entry side **940** of the bathtub **910** and into the bathtub **910**; while continuing to grip one of the first support **120f** and the second support **130f**, steps again over the entry side of the bathtub and into the bathtub **910**. After the user enters the bathtub **910**, the user lets go of the support **120f**, **130f** and the user starts operation of the water controls to start bath/shower. To use the bathtub step stool **900** to exit/get out of the bathtub **910**, a user reaches out and grabs at least one of the supports **120f**, **130f** while standing in the bathtub **910**. The user continues to grip one of the supports **120f**, **130f** and steps one foot laterally out of the bathtub and over the entry side **940**, onto the step assembly **270f** of the bathtub step stool **900**. While continuing to grip one of the supports **120f**, **130f**, the user steps the other foot laterally out of the bathtub and over the entry side **940**, onto the step assembly **270f** of the bathtub step stool **900**. The user then passes between the first support and the second support to step off of the step assembly **270f** and exit the bathtub step stool **900**.

(75) With reference to FIGS. **16A-19**, another embodiment of a step stool **1100** that may be used with a bathtub **910** (or a bed **110**) will be described. Like elements to those shown and/or described with respect to FIGS. **1-15** are shown and/or described with like reference numbers and/or with a “g” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-15** incorporated herein.

(76) Attachments **1120** (e.g., J-shaped members that raise, lower) that go on the back of the bath step stool **1100** (e.g., attached to the leg **160g** of the first support/guard rail **120g** and attached to the

leg **170g** of the second support/guard rail **130g**) terminate at mount assemblies **1130**, which mount onto/over a side/wall **940** of the bathtub **910** as shown in FIG. **16A**. The mount assemblies **1130** each include a bracket **1150** with a downwardly facing U-shaped clamp assembly **1160**. The U-shaped clamp assembly **1160** includes a first stationary clamp member **1170** and a second movable clamp member **1180**. Inner surfaces of the first stationary clamp member **1170** and the second movable clamp member **1180** include respective pads **1190** to prevent scratching of the side/wall **940** of the bathtub **910** when clamped onto the side/wall **940**. Rotating member **1200** is rotatably received in the clamp assembly **1160** and coupled to the second movable clamp member **1180** whereby rotation of the rotating member **1200** causes the second movable clamp member **1180** to move towards/away from the first stationary clamp member **1170** to tighten/loosen the clamp assembly **1160** on the side/wall **940**.

(77) With reference to FIGS. **16A**, **18**, and **19**, the step stool **1100**, which in an alternative embodiment is a bed step stool (e.g., bed step stool **100**, **700**, **849**, **859**, **869**, **879** with folding/collapsible tall grab bar and short grab bar), bath step stool (e.g., bath step stool **900**), or step stool for another application, includes folding/hinge mechanisms **1210** that allow the rails **120g**, **130g** to fold relative to the step assembly **270g** between the deployed step stool **1100** shown in FIG. **16A** and the folded, flat, compact configuration shown in FIGS. **18**, **19**. The folding mechanisms **1210** include pivots **1220**, brackets **1230** connecting end portions **1240** of the step assembly **270g** at the pivots **1220** to lower portions of the rails **120g**, **130g** at fastener locations **1242**, springs **1250**, grip bars **1260**, and c-shaped ends **1270** of the step assembly **270g**. In use, when the step stool **1100** is in the configuration shown in FIG. **16**, to fold the step stool **1100**, a user pulls/grabs the grip bar(s) **1260** toward step crossbar assemblies **190g**, **200g**, against the biasing force of the springs **1250**. This disengages the grip bar(s) **1260** from the c-shaped ends **1270** of the step assembly **270g**, allowing the folding mechanisms **1210** and the rails **120g**, **130g** to pivot about the pivots **1220** to the flat, compact configuration of FIGS. **18**, **19** for transporting/storage. To deploy the step stool **1100**, the folding mechanisms **1210** and the rails **120g**, **130g** are pivoted from the flat, compact configuration of FIGS. **18**, **19** about the pivots **1220** to the configuration shown in FIG. **16**. The springs **1250** urge the grip bar(s) **1260** into the c-shaped ends **1270**, locking the rails **120g**, **130g** in a perpendicular orientation relative to the step assembly **270g**. Another advantage of the folding/hinge mechanisms **1210** is that they allow for an already assembled product to the consumer where the arms just pop up and do not require fasteners for assembly.

(78) As shown in FIG. **16**, the step stool **1100** or other step stool (e.g., step stool **100**, **700**, **849**, **859**, **869**, **879**, **900**) includes a second motion-detecting lighting assembly **1280** for illuminating via one or more lights (e.g., LEDs) the area in front of the step stool **1100** upon detected motion (e.g., movement of the user's legs) of the user by a motion sensor. The second motion-detecting lighting assembly **1280** is flush with a front of the step stool **1100**.

(79) As shown in FIG. **16B**, the step stool **1100** or other step stool (e.g., step stool **100**, **700**, **849**, **859**, **869**, **879**, **900**) includes glow-in-the-dark strips **1290** with phosphorescent pigment (e.g., zinc sulfide) at one or more of locations **1292**, **1294**, **1296**, **1298**, **1300**, **1302**, **1304** to help the user know where the edges of the step stool **1100** are so the user can see easier at night.

(80) The step stool **1100** or other step stool (e.g., step stool **100**, **700**, **849**, **859**, **869**, **879**, **900**) includes foam covers **1320** that cover lower portions **1330** of the front legs **250g**, **260g**, protecting any corners/edges and helping to prevent the user from scraping their foot and getting hurt.

(81) The step stool **1100** or other step stool (e.g., step stool **100**, **700**, **849**, **859**, **869**, **879**, **900**) may come in different colors.

(82) The step stool **1100** or other step stool (e.g., step stool **100**, **700**, **849**, **859**, **869**, **879**, **900**) may include a non-slip, warmer temperature, more comfortable upper surface material **1340** on the step assembly **270g**.

(83) Step assembly **270g** may be vertically adjustable with respect to guard rails **120g**, **130g**. In the embodiment of the step stool **1100**, the guard rails **120g**, **130g** are of the same height and may have

rubberized non slip foot/rubber tip **1012g** at bottom of guard rails **120g**, **130g** and/or rubber sleeves **1350** at top of guard rails **120g**, **130g** for secure gripping of guard rails **120g**, **130g** by the user when transferring to and from the bathtub or shower.

(84) With reference to FIG. **20**, another embodiment of a bed step stool **1400** will be described, with like elements to those shown and/or described with respect to FIGS. **1-4** shown and/or described with like reference numbers, but with a “h” suffix, and with the subject matter described above with respect to FIGS. **1-4** incorporated herein. A ramp **1410** is pivotally attached to a front **1420** of the step assembly **270h**. The ramp **1410** may allow some users who are not able to take a larger step (e.g., users that shuffle their feet to walk) and use a wheeled support device (e.g., wheelchair) easier access up to the step assembly **270h**. When not in use, the ramp **1420** may just under the step assembly **270h** so that the ramp **1410** is parallel to and underneath the step assembly **270h**. Alternatively, the ramp **1410** may pivotally upwardly 180+ degrees so that an upper side **1430** of the ramp **1410** rests on an upper side **1440** of the step assembly **270h**.

(85) With reference to FIG. **21**, an additional embodiment of a bed step stool **1500** will be described, with like elements to those shown and/or described with respect to FIGS. **1-4** shown and/or described with like reference numbers, but with an “i” suffix, and with the subject matter described above with respect to FIGS. **1-4** incorporated herein. The step assembly **270i** includes a wider front edge **1510**, where a user steps onto the step assembly **270i**, and a narrower rear edge **1520** next to the bed. As a result, a distance F between the front legs **250i**, **260i** is greater than a distance B between the rear legs **160i**, **170i**. This design allows a user a wider area to access the bed step stool **1500** while funneling or focusing the user's travel across the step assembly **270i** to a more focused location, which may be important for having the user enter and exit the bed at a certain desired location to reduce the user having to move (or having to move the user) towards or away from the head of the bed once the user is on the bed.

(86) With reference to FIGS. **22A** and **22B**, further embodiments of bed step stools **1600**, **1700** will be described, with like elements to those shown and/or described with respect to FIGS. **1-4** shown and/or described with like reference numbers, but with “j” and “k” suffices, and with the subject matter described above with respect to FIGS. **1-4** incorporated herein. The bed step stool **1600** of FIG. **22A** includes a pair of step assemblies **1610**, **1620** at different heights/locations, enabling a user that is not able to take larger steps to take smaller steps to get up to and on or off the bed. The bed step stool **1600** includes additional central vertical supports **1630** and frame supports **1640** to support a second step assembly **1620**. The bed step stool **1700** of FIG. **22B** is similar to the bed step stool **1600** of FIG. **22A**, but includes three step assemblies **1710**, **1720**, **1730** at different heights/locations, enabling a user that is not able to take larger steps to take even smaller steps to get up to and on or off the bed. Similar to the bed step stool **1600** of FIG. **22A**, the bed step stool **1700** of FIG. **22B** includes additional central vertical supports **1740** and frame supports **1750**, **1760** to support a second step assembly **1720** and a third step assembly **1730**.

(87) With reference to FIGS. **23-29**, an additional embodiment of a bed step stool **1800**, which may also function as a walker, will be described. Like elements to those shown and/or described with respect to FIGS. **1-4** are shown and/or described with like reference numbers, but with a “l” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-4** incorporated herein. The bed step stool/walker **1800** is shown adjacent to bed **1101**.

(88) The bed step stool/walker **1800** includes guard rails **1201**, **130L** that are shorter in height than guard rails **120**, **130** of FIGS. **1-4**, step crossbar assemblies **190I**, **200I**, **1810**, **1820** that connect back legs **160I**, **170I** to front legs **250I**, **260I**, and upper cross bars **1830**, **1840**. Wheels **1850** are disposed at lower portions **1860** of legs **160I**, **170I**, **250I**, **260I**. A brake mechanism **1870** is operably coupled to the wheels **1850**. A step assembly **1880** is supported by the step crossbar assemblies **190I**, **200I**, **1810**, **1820**. A seat assembly **1890** with a seat **1900** is rotatably movably coupled at pivot points **1910** to the back legs **160I**, **170I** for rotation/movement between a stored position (FIGS. **23**, **25**) and a deployed seat position (FIG. **26-29**). The front legs **250I**, **260I**

terminate in an upper portion **1920** in handles **1930**, which include hand brakes **1940** operably coupled to the brake mechanism **1870**. The handles **1930**/hand brakes **1940** may be disposed in a direction away from the seat assembly **1890** or in a direction toward the seat assembly **1890**. The front legs **250I**, **260I** include upper and lower height adjustment mechanisms **1950**, **1960**. The back legs **160I**, **170I** include height adjustment mechanisms **1970**.

(89) In use, the bed step stool/walker **1800** may be used as a walker and a seat when the seat assembly **1890** is disposed in the deployed seat position. The bed step stool/walker **1800** is rolled over next to the bed (or other object to be stepped up to) **1101** and the seat assembly **1890** is rotatably moved to the stored position. The user then actuates the brake mechanism **1870** to secure the bed step stool/walker **1800** into position and steps onto the step assembly **1880** to assist in getting onto and/or off of the bed **1101** in the manner described above with respect to FIGS. **1-4**, which is incorporated herein. The handles **1930** may be gripped for additional assistance/security when stepping onto and/or off of the step assembly **1880**.

(90) In one or more embodiments, the walker **1800** does not function as and/or is not used/designed for used as a bed step stool.

(91) With reference to FIGS. **30-35**, an additional embodiment of a bed step stool **2000**, which may also function as a walker, will be described. Like elements to those shown and/or described with respect to FIGS. **1-4** and FIGS. **23-29** are shown and/or described with like reference numbers, but with a “m” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-4** and FIGS. **23-29** incorporated herein. The bed step stool/walker **2000** is shown adjacent to bed **110m**. The bed step stool/walker **2000** is similar to the bed step stool/walker **1800**, but instead of seat assembly **1890** coupled to the back legs **160I**, **170I**, preventing a user to walk through bed step stool/walker **1800** from one side/front, onto/over step assembly **1880m**, to opposite side/rear, the bed step stool **2000** includes a seat assembly **1890m** that is movable between a stored position (FIGS. **30**, **32**), providing an open path for a user to walk through bed step stool/walker **2000** from one side/front **2010**, onto/over step assembly **1880m**, to opposite side/rear **2020**, and a deployed seat position (FIGS. **31**, **33-35**), blocking the path. In the stored position (FIGS. **30**, **32**), the seat assembly **1890m**, which includes couplers **2030** on a bottom **2040**, is coupled to one of the upper cross bars so that the seat **1900m** is disposed along an outer side **2050** of the bed step stool/walker **2000**. With the seat assembly **1890m** in the stored position, the upper cross bars **1830m**, **1840m** and handles **1930m** form first support(s) and second support(s) that define an unobstructed pathway for entry and exit with respect to the bed **110m** and the bed step stool/walker **2000** that is perpendicular with respect to the sides of the bed **110m** whereby the user goes between the first support(s) **1830m**, **1930m** and the second support(s) **1840m**, **1930m**, onto the step assembly **1880m**, and off the step assembly **1880m** without any obstruction. In the deployed seat position (FIGS. **31**, **33-35**), the seat assembly **1890m** straddles (and is coupled via the couplers **2030** to) the upper cross bars **1830m**, **1840m** and obstructs the pathway when used to assist the user into/out of bed **110m**.

(92) In one or more embodiments, the walker **2000** does not function as and/or is not used/designed for used as a bed step stool.

(93) With reference to FIGS. **36** and **37**, another embodiment of a bed step stool **2100**, which may also function as a walker, will be described. Like elements to those shown and/or described with respect to FIGS. **1-4** and FIGS. **23-35** are shown and/or described with like reference numbers, but with a “n” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-4** and FIGS. **23-35** incorporated herein. The bed step stool/walker **2100** is similar to the bed step stool/walker **2000**, but instead of seat assembly **1890m** removably and movably coupled to one of the upper cross bars **1830m**, **1840m** as shown in FIG. **30**, seat assembly **1890n** is rotatably coupled to one of upper cross bars **1830n**, **1840n** for rotation between a stored position (FIG. **36**) along outer side **2050n**, providing an open path for a user to walk through bed step stool/walker **2100** from one side/front **2010n**, onto/over step assembly **1880n**, to opposite side/rear **2020n**, and a deployed seat position (FIG. **37**), blocking the path. In the stored position (FIG. **36**), the seat

assembly **1890n**, which includes rotatable couplers **2110** on a bottom **2040n**, is coupled to one of the upper cross bars **1830n**, **1840n** so that the seat **1900n** is disposed along the outer side **2050n** of the bed step stool/walker **2100**. With the seat assembly **1890n** in the stored position, the upper cross bars **1830n**, **1840n** and handles **1930n** form first support(s) and second support(s) that define an unobstructed pathway for entry and exit with respect to the bed **110m** and the bed step stool/walker **2100** that is perpendicular with respect to the sides of the bed **110m** whereby the user goes between the first support(s) **1830n**, **1930n** and the second support(s) **1840n**, **1930n**, onto the step assembly **1880n**, and off the step assembly **1880n** without any obstruction. In the deployed seat position (FIG. 37), the seat assembly **1890n** straddles and is supported by the upper cross bars **1830n**, **1840n** and obstructs the pathway when used to assist the user into/out of bed **110m**.

(94) In one or more embodiments, the walker **2100** does not function as and/or is not used/designed for used as a bed step stool.

(95) With reference to FIGS. 38 and 39, another embodiment of a bed step stool **2200**, which may also function as a walker, will be described. Like elements to those shown and/or described with respect to FIGS. 1-4 and FIGS. 23-37 are shown and/or described with like reference numbers, but with an “o” suffix, and with the subject matter shown and/or described above with respect to FIGS. 1-4 and FIGS. 23-37 incorporated herein. The bed step stool/walker **2200** is similar to the bed step stool/walkers **2000**, **2100**, but additionally includes L-shaped upper handrails **2210** and a guard rail/back support **2220** rotatably movably coupled to one of the L-shaped upper handrails **2210** via couplers **2230**. The guard rail/back support **2220** is movable and rotatable between a stored position or out-of-the-way position (FIG. 39) along outer side **2050o**, providing an open path for a user to walk through bed step stool/walker **2200** (when seat assembly **1890o** is also in stored position) from one side/front **2010o**, onto/over step assembly **1880o**, to opposite side/rear **2020o**, and a deployed safety position (FIG. 38), providing back support, preventing the user from falling backward, and blocking the path. With the seat assembly **1890o** and the guard rail/back support **2220** in the stored position or out-of-the-way position, the upper cross bars **1830o**, **1840o** and handles **1930o** form first support(s) and second support(s) that define an unobstructed pathway for entry and exit with respect to the bed **110m** and the bed step stool/walker **2200** that is perpendicular with respect to the sides of the bed **110m** whereby the user goes between the first support(s) **1830o**, **1930o** and the second support(s) **1840o**, **1930o**, onto the step assembly **1880o**, and off the step assembly **1880o** without any obstruction. In the deployed position(s) (FIG. 38), the seat assembly **1890o**, which straddles the second support(s) **1840o**, **1930o**, and the guard rail/back support **2220** obstruct the pathway when used to assist the user into/out of bed **110m**.

(96) In one or more embodiments, the walker **2200** does not function as and/or is not used/designed for used as a bed step stool.

(97) With reference to FIG. 40, an additional embodiment of a bed step stool **2300**, which may be wheeled from location to location, will be described. Like elements to those shown and described with respect to FIGS. 1-4 and FIGS. 23-39 are shown and/or described with like reference numbers, but with a “p” suffix, and with the subject matter shown and/or described above with respect to FIGS. 1-4 and FIGS. 23-39 incorporated herein. The bed step stool **2300** is generally similar to the bed step stool **100**, but includes a wheel assembly **2310** including one or more wheels **2320** coupled to the step assembly **270p** at a side **2330** of the step assembly **270p** via wheel clamp/mount **2340** and step crossbar assembly **200p** (or step crossbar assembly **190p**), or may be coupled to the step assembly **270p** via wheel mount **2340** and legs **280p**. The wheel assembly **2310** may be detachably coupled to the side **2330** of the bed step stool **2300**.

(98) In use, the bed step stool **2300** is wheeled from location to location (e.g., from storage to side of bed, from side of bed to storage) by pulling on the upper portion **440p** of the tall guardrail **120p** to cause the bed step stool **2300** to tilt/lean about the pair of rubber tips **400p** of the tall guardrail **120p** towards the side **2330** in the direction of the arrow shown until the bed step stool **2300** is supported by the wheel assembly **2310**. This orientation is a tilted/leaned/wheel assembly

supported orientation. Using the upper portion **440p** of the tall guardrail **120p** as a handle bar gripped by one's hands, the bed step stool **2300** is pushed or pulled (and rolled) using the wheel assembly **2310** like a cart to the desired location. At the desired location, the bed step stool **2300** is positioned for use by handling the upper portion **440p** and initially tilting/leaning the bed step stool **2300** about the wheel assembly **2310** until the pair of rubber tips **400p** of the tall guardrail **120p** engage the support surface/floor and then the bed step stool **2300** is continued to be tilted about the rubber tips **400p** until the bed step stool **2300** is disposed in position in an orientation similar to those shown in FIGS. **2** and **40** and used in the manner described herein. This bed step stool orientation is a stabilized/wheel assembly out of the way orientation.

(99) With reference to FIG. **41**, another embodiment of a bed step stool **2400**, which includes a step assembly **270q** that may be raised and lowered, will be described. Like elements to those shown and described with respect to FIGS. **1-4** and FIGS. **23-40** are shown and/or described with like reference numbers, but with a “q” suffix, and with the subject matter shown and/or described above with respect to FIGS. **1-4** and FIGS. **23-40** incorporated herein. A raising and lowering mechanism **2410** is disposed under the step assembly **270q** and is used to raise and lower the step assembly **270q** (or allows the raising and/or lowering of the step assembly **270q**) for users that may have more difficult time stepping onto and/or stepping off of the step assembly **270q** shown and described herein. The raising and lowering mechanism **2410** may be, for example, a stair stepper mechanism **2420** that depresses/lowers when one steps on the step assembly **270q** and then recoils, lifting the user up after both feet are on the step assembly **270q**. Alternatively, the raising and lowering mechanism **2410** may be a scissor hydraulic (or pneumatic/scissor pneumatic) vertical lift mechanism **2430** that a user controls via manual control **2440** to raise/lower the step assembly **270q** as desired. Alternatively, the raising and lowering mechanism **2410** may be a counterbalance vertical lift mechanism **2450** that a user controls via manual control **2440** to raise/lower the step assembly **270q** as desired.

(100) In use, the user either steps on the step assembly **270q** (e.g., when using the stair stepper mechanism **2420**) or the user lowers the step assembly **270q** (via the scissor hydraulic (or pneumatic/scissor pneumatic) vertical lift mechanism **2430** or the counterbalance vertical lift mechanism **2450**) using the manual control **2440** until the step assembly **270q** is at a suitable height for the user to step thereon. The user then steps on the step assembly **270q**, and the raising and lowering mechanism **2410** automatically raises (e.g., via the stair stepper mechanism **2420**) the step assembly **270q** and user to a desired height or the user uses the manual control **2440** (via the scissor hydraulic (or pneumatic/scissor pneumatic) vertical lift mechanism **2430** or the counterbalance vertical lift mechanism **2450**) to raise the step assembly **270q** and user to the desired height. When getting off the bed step stool **2400**, the user steps on the step assembly **270q** and the step assembly **270q** automatically slowly lowers (e.g., via the stair stepper mechanism **242**) until it is adjacent to/on the support surface/ground, or the user uses the manual control **2440** to lower the step assembly **270q** and user (via the scissor hydraulic (or pneumatic/scissor pneumatic) vertical lift mechanism **2430** or the counterbalance vertical lift mechanism **2450**) to the desired height such as adjacent to/on the support surface/ground. The user then steps off the step assembly **270q** and onto the support surface/floor. Thus, the bed step stool **2400** includes a first orientation where the raising and lowering mechanism orients the step assembly in a raised position to facilitate getting the user into and out of bed and a second orientation where the raising and lowering mechanism orients the step assembly in a lowered position to facilitate getting the user onto and off of the step relative to a support surface.

(101) The above figures may depict exemplary configurations for the invention, which is done to aid in understanding the features and functionality that can be included in the invention. The invention is not restricted to the illustrated architectures or configurations, but can be implemented using a variety of alternative architectures and configurations. Additionally, although the invention is described above in terms of various exemplary embodiments and implementations, it should be

understood that the various features and functionality described in one or more of the individual embodiments with which they are described, but instead can be applied, alone or in some combination, to one or more of the other embodiments of the invention, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus the breadth and scope of the present invention, especially in the following claims, should not be limited by any of the above-described exemplary embodiments. (102) Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as mean “including, without limitation” or the like; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; and adjectives such as “conventional,” “traditional,” “standard,” “known” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. Likewise, a group of items linked with the conjunction “and” should not be read as requiring that each and every one of those items be present in the grouping, but rather should be read as “and/or” unless expressly stated otherwise. Similarly, a group of items linked with the conjunction “or” should not be read as requiring mutual exclusivity among that group, but rather should also be read as “and/or” unless expressly stated otherwise. Furthermore, although item, elements or components of the disclosure may be described or claimed in the singular, the plural is contemplated to be within the scope thereof unless limitation to the singular is explicitly stated. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent.

Claims

1. A bed step stool for assisting a user into and out of bed including opposite sides, a head, and a foot, comprising: a first support to be disposed closer to the head of the bed; a second support to be disposed closer to the foot of the bed, the bed step stool including a side and at least one of the first support and the second support including an upper portion configured to be gripped by a user; a wheel assembly coupled to the side of the bed step stool and including one or more wheels configured to roll on a surface; a step assembly supported between the first support and the second support, the step assembly to enable the user to step thereon for assisting the user into and out of the bed; wherein the first support and the second support define an unobstructed pathway for entry and exit with respect to the bed and the bed step stool that is perpendicular with respect to the sides of the bed whereby the user goes between the first support and the second support, onto the step assembly, and off the step assembly without any obstruction, and the bed step stool including a first orientation where the bed step stool is stabilized on the surface and the wheel assembly coupled to the side of the bed step stool is out of the way, and a second orientation where the bed step stool is tilted towards the side and is supported by the wheel assembly coupled to the side of the bed step stool, allowing rolling movement of the bed step stool on the surface via the one or more wheels of the wheel assembly.
2. The bed step stool of claim 1, wherein the wheel assembly is detachably coupled to the side of the bed step stool.
3. A bed step stool for assisting a user into and out of bed including opposite sides, a head, and a foot, comprising: a first support to be disposed closer to the head of the bed; a second support to be disposed closer to the foot of the bed; a wheel assembly coupled to the side of the bed step stool and including one or more wheels configured to roll on a surface; a step assembly supported between the first support and the second support, the step assembly to enable the user to step

thereon for assisting the user into and out of the bed; a raising and lowering mechanism disposed under the step assembly configured to raise and lower the step assembly; wherein the first support and the second support define an unobstructed pathway for entry and exit with respect to the bed and the bed step stool that is perpendicular with respect to the sides of the bed whereby the user goes between the first support and the second support, onto the step assembly, and off the step assembly without any obstruction, and the bed step stool including a first orientation where the raising and lowering mechanism orients the step assembly in a raised position to facilitate getting the user into and out of bed and a second orientation where the raising and lowering mechanism orients the step assembly in a lowered position to facilitate getting the user onto and off of the step relative to a support surface.

4. The bed step stool of claim 3, wherein the raising and lowering mechanism is a stair stepper mechanism that lowers when the user steps on the step assembly and then recoils, lifting the user up after both feet are on the step assembly.

5. The bed step stool of claim 3, wherein the raising and lowering mechanism is a manual control and a hydraulic or pneumatic vertical lift mechanism that a user controls via the manual control to raise and lower the step assembly to a desired height.

6. The bed step stool of claim 3, wherein the raising and lowering mechanism is a manual control and a counterbalance vertical lift mechanism that a user controls via the manual control to raise and lower the step assembly to a desired height.
